# FINAL DATA ANALYTICS PROJECT

Zomato Delivery Performance Analysis

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| **Organization** | SWYNEX Technologies Pvt. Ltd. |
| **Project type** | Data cleaning, exploratory analysis and Power BI dashboard |
| **Tools** | Python, Pandas, Jupyter Notebook, Power BI |

This case study combines the project work completed across data preparation, exploratory data analysis, and interactive dashboarding. It summarizes the business question, approach, major findings, and practical recommendations.

Submission document • Fresher-level project report

# 1. Executive Summary

This project examines food-delivery order records to understand delivery duration and how it varies with traffic density, weather, vehicle type, city category, festivals, and time. The work combines data cleaning and preparation, Python-based exploratory data analysis (EDA), and a Power BI dashboard for interactive exploration.

The dataset used for the analysis contains 45,584 rows and 20 columns after preparation. The overall average delivery time is approximately 26.29 minutes, with a median of 26 minutes. The dashboard provides KPI cards, category comparisons, monthly order counts, and slicers to explore subsets of the data.

# 2. Problem Statement and Objectives

**Problem statement:** Delivery duration can be influenced by operational and environmental conditions. The goal is to explore the available order data and identify patterns that may help explain differences in delivery time.

**Objectives:**

• Prepare a consistent dataset for analysis.
• Explore delivery time against traffic, weather, vehicle type, city category, festival status, and month.
• Build a Power BI dashboard that allows users to compare key categories interactively.
• Communicate findings and recommendations while noting data limitations.

# 3. Dataset Overview

The project uses a Zomato food-delivery dataset with order-level fields such as order date, delivery time, traffic density, weather conditions, vehicle type, city category, and festival status. The prepared file used in the analysis is named **zomato_cleaned_by_manisha.csv**.

Prepared dataset size: **45,584 rows × 20 columns**. Delivery duration is measured in minutes. Missing or unknown category values were retained where appropriate and should be treated cautiously during interpretation.

# 4. Data Cleaning and Preparation

The cleaning stage prepared the data for reliable analysis and dashboard use. The workflow included checking data quality, handling missing or unknown category values, standardizing fields for analysis, and saving a cleaned dataset for downstream work. The final cleaning decisions should be described consistently with the submitted Task 1 notebook or repository.

A useful reporting practice is to preserve unknown categories when their original value cannot be established, rather than guessing a replacement. In this project, weather and traffic include unknown values, so category averages should be read with that limitation in mind.

# 5. Exploratory Data Analysis

Python EDA was used to summarize delivery time and compare patterns across important dimensions. The following figures are based on the recorded project results:

| Metric / segment | Result | Interpretation |
|---|---|---|
| Total prepared records | 45,584 | Number of records in the cleaned dataset |
| Average delivery time | 26.29 min | Overall mean delivery duration |
| Median delivery time | 26 min | Half of recorded deliveries are at or below this value |
| Minimum / maximum | 10 / 54 min | Observed range in the prepared dataset |
| Standard deviation | 9.38 min | Delivery duration varies around the mean |
| Traffic: Jam | 14,139 orders; 31.18 min avg | Highest average among listed traffic categories |

| Metric / segment | Result | Interpretation |
|---|---|---|
| Traffic: Low | 15,476 orders; 21.27 min avg | Lower average than jam conditions |
| Weather: Sunny | 21.86 min avg | Lowest average among listed weather categories |
| Weather: Cloudy / Fog | 28.92 / 28.91 min avg | Higher averages among listed weather categories |
| Festival = Yes | 896 orders; 45.52 min avg | Festival orders show a notably higher mean |
| Festival = No | 44,460 orders; 25.98 min avg | Non-festival orders form the majority |

These are descriptive associations, not proof that a factor alone causes slower delivery. Order mix, distance, time of day, and other variables may also influence the results.

# 6. Power BI Dashboard

The Power BI report is titled **Zomato Delivery Performance Dashboard**. It includes KPI cards for total orders and average delivery time, visuals comparing delivery time by traffic density and weather, city-category order counts, vehicle-type comparisons, order-type counts, monthly order counts, and slicers for traffic density, weather, and festival status.

The slicers allow a reviewer to filter the report and inspect how selected conditions affect the displayed metrics. The screenshots below show the dashboard as captured in Power BI Desktop. Note that the screenshots show selected filters, so the visible KPI values are filtered values rather than the overall dataset totals.

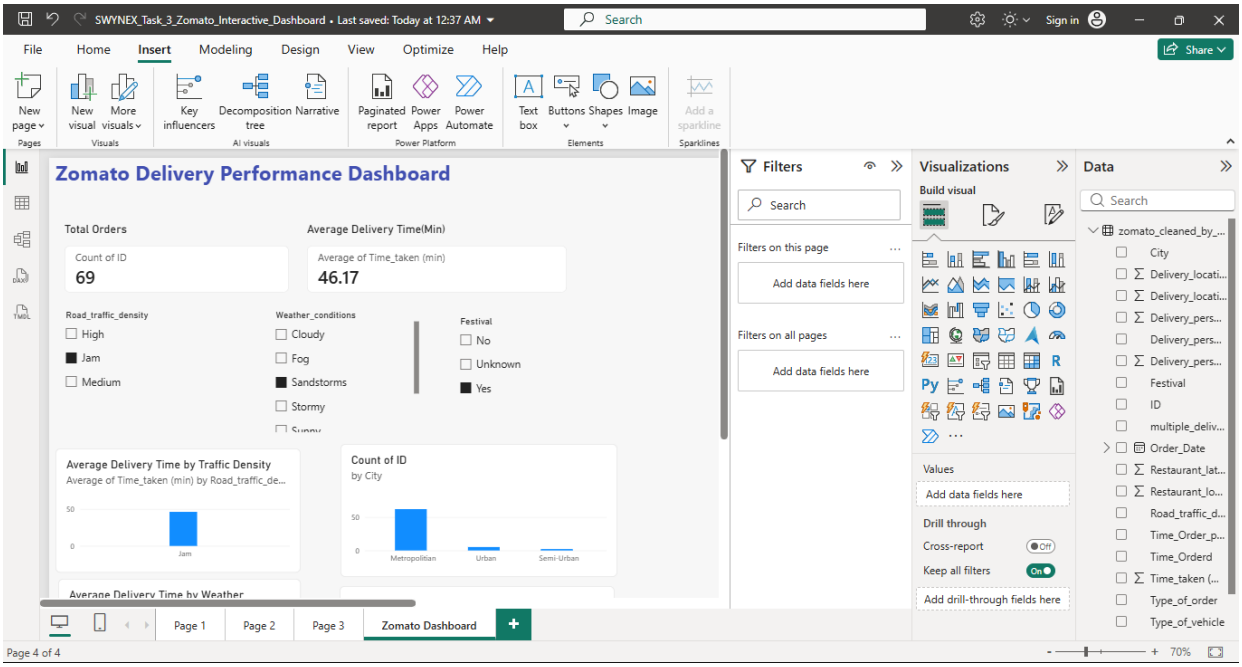



Figure 1. Dashboard overview with KPI cards and slicers visible.

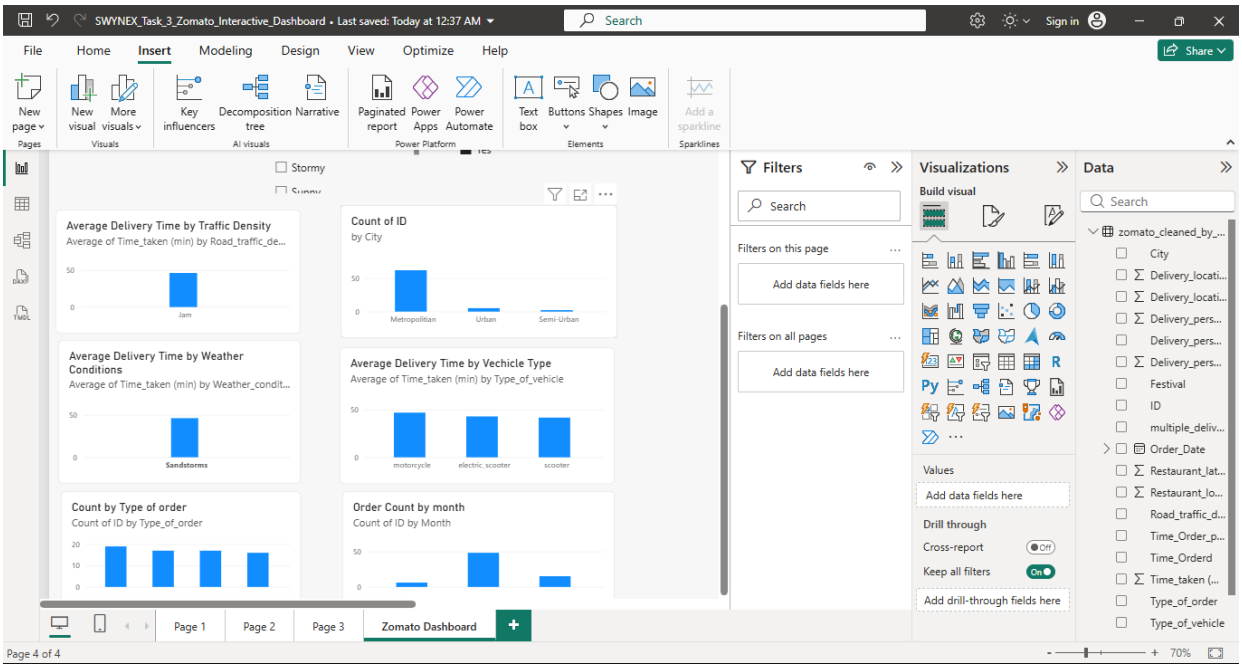



Figure 2. Lower dashboard section showing category comparisons and monthly order counts.

# 7. Key Findings

• Traffic conditions show a meaningful descriptive difference: the jam category has a 31.18-minute average, compared with 21.27 minutes for low traffic.
• Weather categories differ in average delivery time. Sunny conditions average 21.86 minutes, while cloudy and foggy conditions average about 28.9 minutes.
• Orders marked as festival-related have a much higher average delivery time (45.52 minutes) than orders marked No (25.98 minutes); however, the festival-Yes group is much smaller.
• Motorcycle deliveries average 27.61 minutes, while scooter and electric-scooter deliveries average about 24.5 minutes. This is an observed difference and should not be interpreted as a vehicle-performance conclusion without considering delivery distance and order mix.
• Most records fall under the Metropolitian city category (34,087), followed by Urban (10,133). Category naming and source-data quality should be verified before presenting this as a geographic conclusion.

# 8. Recommendations

1. **Plan for peak traffic:** Review dispatch allocation and expected delivery-time communication for high-traffic periods.
2. **Prepare for festival demand:** Compare order volume and delivery duration during festival periods and consider staffing or rider availability adjustments if the pattern is consistent.
3. **Monitor weather-sensitive operations:** Track weather and delivery duration over a larger time window before making operational changes.
4. **Use the dashboard for monitoring:** Review KPI changes after filtering by traffic, weather, and festival status.
5. **Improve data quality:** Validate unknown categories, inconsistent labels, and unusual values before making high-impact decisions.

# 9. Limitations

• The analysis is descriptive and does not establish causality.
• Unknown values remain in some categorical fields, including weather and traffic.
• The dataset covers February–April 2022 in the recorded monthly summary; it may not represent current operations or seasonality across a full year.
• Vehicle-condition codes should not be given a business meaning unless their definitions are confirmed from the dataset documentation.
• Some dashboard screenshots show active slicer selections. The values displayed in those screenshots are therefore not necessarily the overall totals.

# 10. Tools and Deliverables

**Tools:** Python, Pandas, Jupyter Notebook, Power BI, GitHub.
**Task 1:** Data cleaning and preparation.
**Task 2:** Exploratory data analysis using Python.
**Task 3:** Interactive Power BI dashboard.
**Task 4:** Combined case-study report with screenshots and links.

# 11. Project Repositories

Task 1 – Data Cleaning & Preparation:
https://github.com/manisha88156/SWYNEX-Data-Cleaning-Preparation

Task 2 – Zomato Exploratory Data Analysis:
https://github.com/manisha88156/SWYNEX-Zomato-Exploratory-Data-Analysis

Task 3 – Zomato Interactive Dashboard:
https://github.com/manisha88156/SWYNEX-Zomato-Interactive-Dashboard

Suggested Task 4 repository name: **SWYNEX-Final-Data-Analytics-Project**. Add this PDF, the dashboard screenshots, and the Power BI **.pbix** file to the final repository, if requested by the task instructions.

# 12. Conclusion

This project demonstrates a complete beginner-level analytics workflow: preparing a dataset, exploring patterns in Python, and presenting results through an interactive Power BI dashboard. The analysis highlights differences in delivery duration across traffic, weather, festival status, and vehicle type. These findings can guide further investigation, while the stated limitations should be considered before operational decisions are made.

**Final submission check:** confirm that the Task 4 repository is public or otherwise accessible to the reviewer; verify the PDF and .pbix upload; include the requested LinkedIn explanation post URL; and test all repository links in a private/incognito browser window.